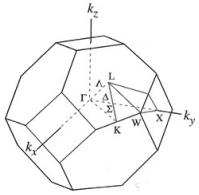
PN Junction



What is a Diode Good for?

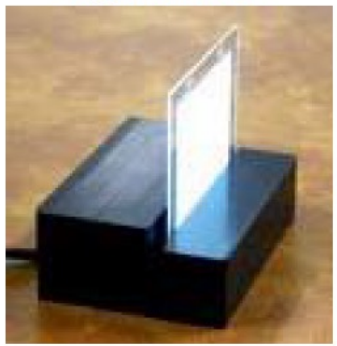
solar cells



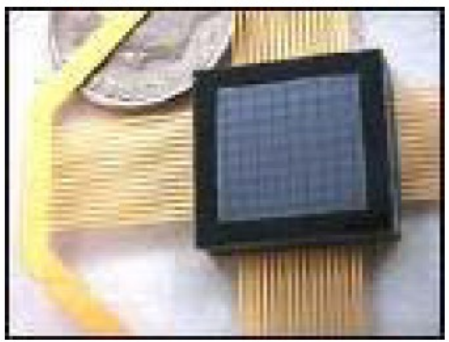
GaAs lasers



Organic LED



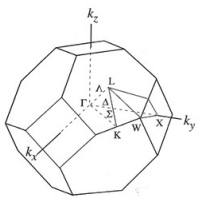
Avalanche Photodiode



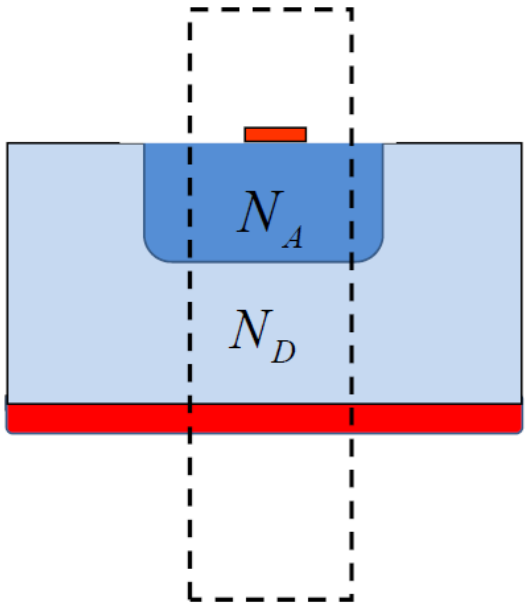
GaN lasers



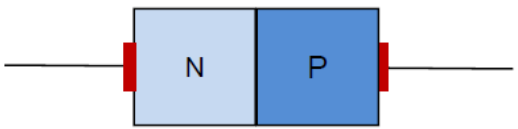
Image.google.com

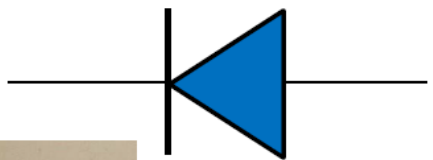


PN Junction Devices

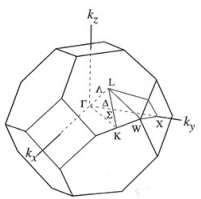


Symbols

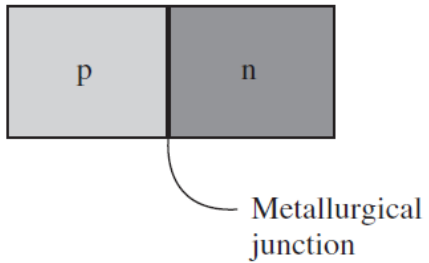




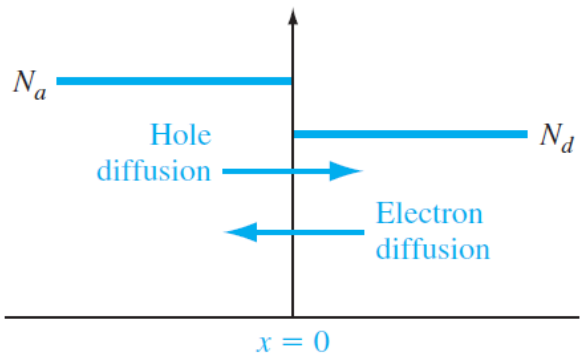




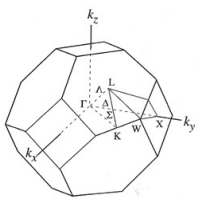
Basic Structure of PN Junction



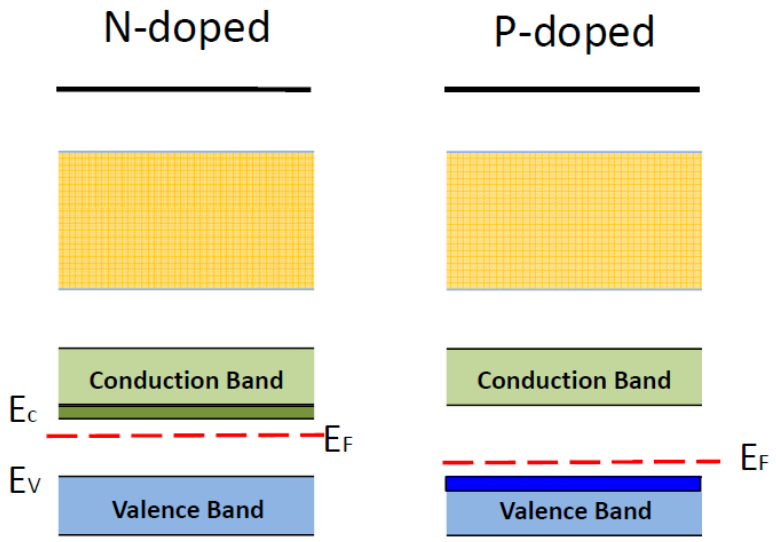
(a)



(b)

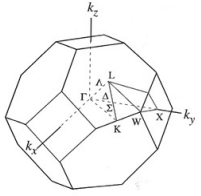


PN Junctions - Equilibrium



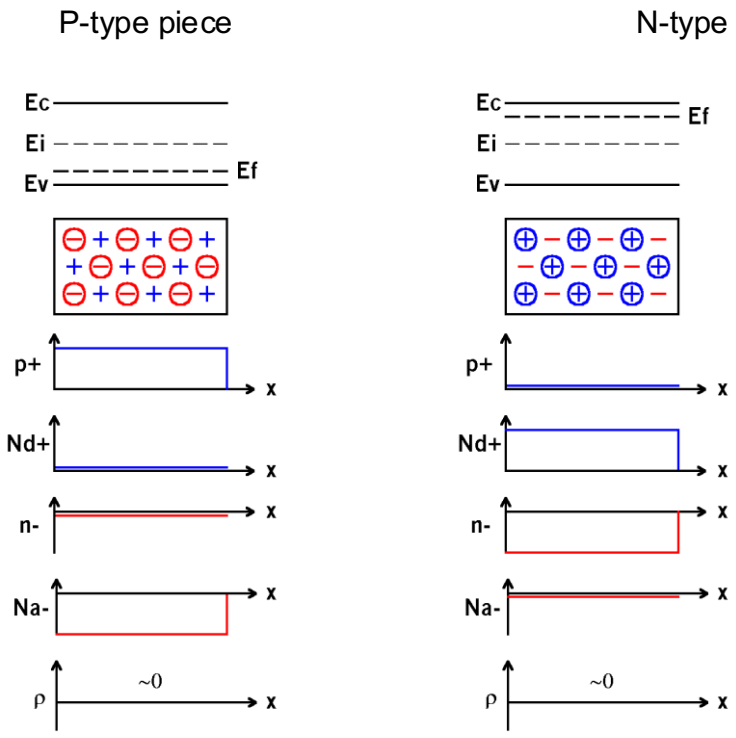
What happens when these band structures collide?

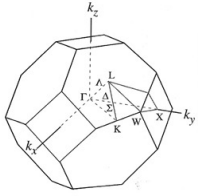
- Fermi energy must be constant at equilibrium, so bands must bend near interface



Time < 0: Pieces Separated

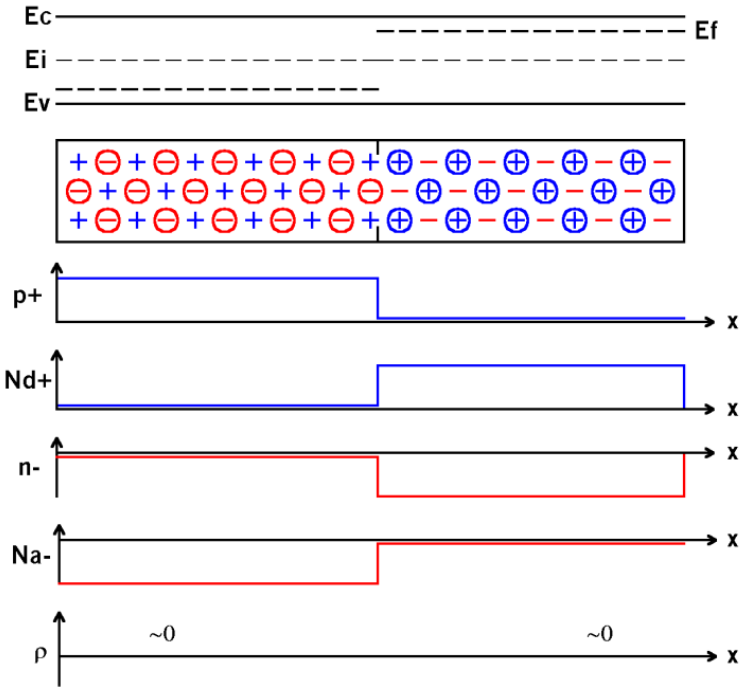
Time < 0

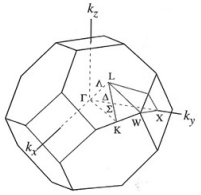




Forming a Junction

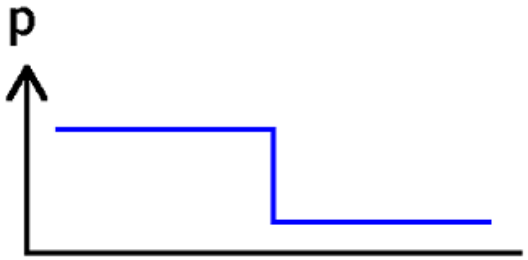
At time = 0, slam the two pieces together



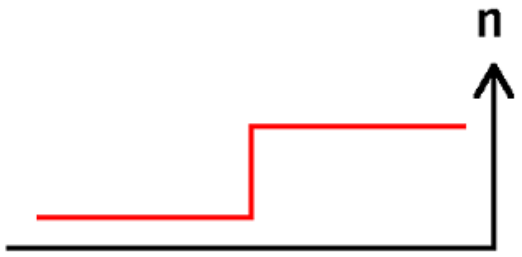


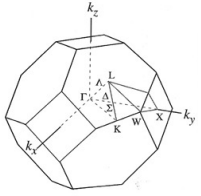
Gradients Drive Diffusion

- Hole gradient



- Electron gradient





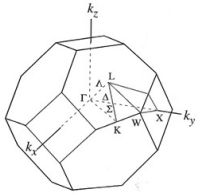
Gradients Drive Diffusion



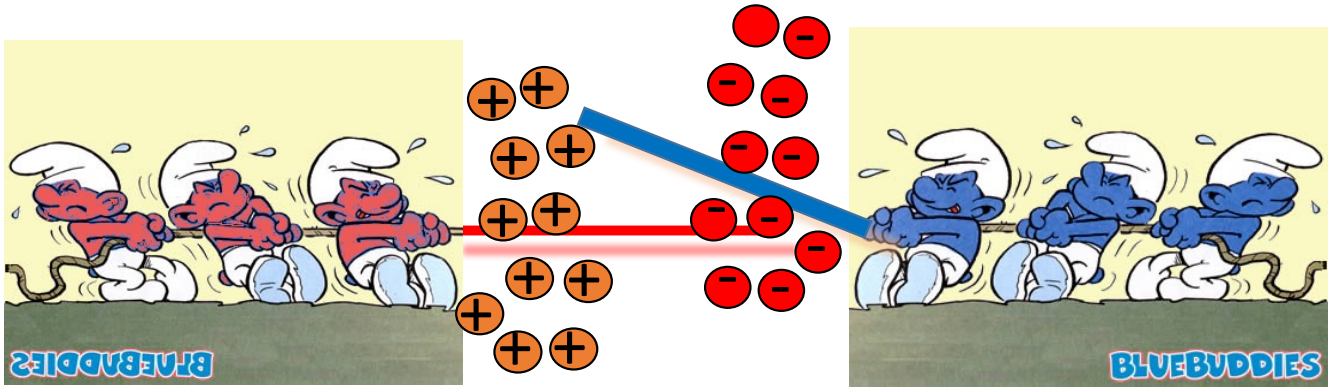


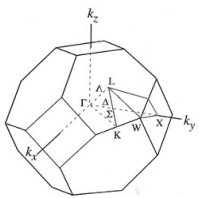




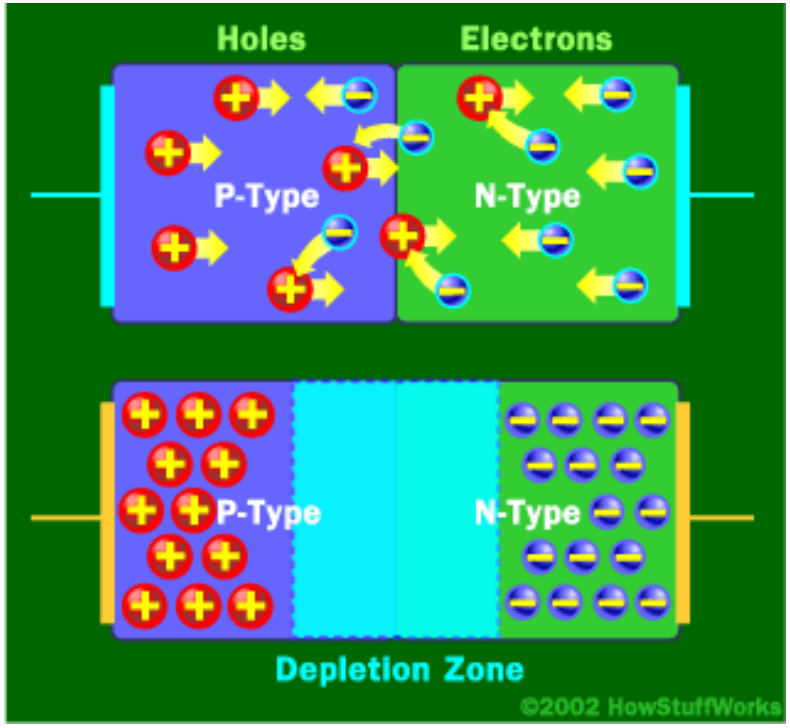


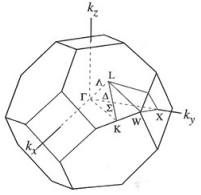
- But charges can’t venture too far from the interface because their Coulomb forces pull them back!





- Separation of a sea of charge, leaving behind a charge depleted region

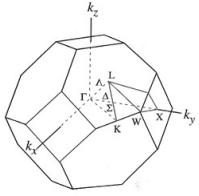




- Separation of a sea of charge, leaving behind a charge depleted region



http://scott.club365.net/uploaded_images/Moses-Parts-the-Red-Sea-2-782619.jpg



Built-in Potential Barrier

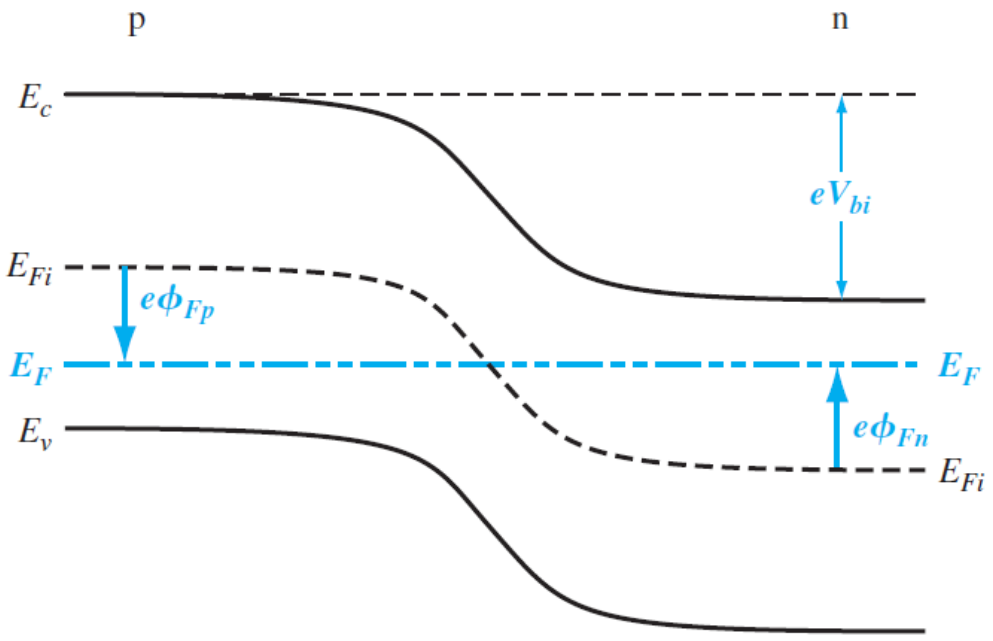
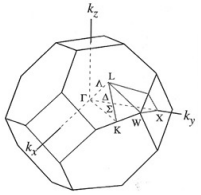


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.



Built-in Potential Barrier

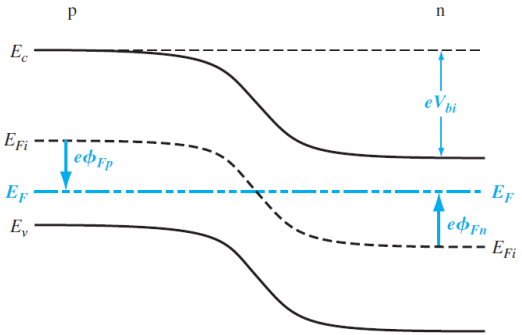


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

Two Assumptions:

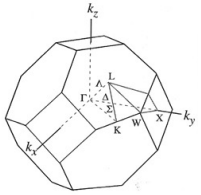
- Boltzmann Approximation
- Complete Ionization

$$V_{bi} = |\emptyset_{Fn}| + |\emptyset_{Fp}|$$

$$\emptyset_{Fn} = \frac{-kT}{e}\ln\left(\frac{N_d}{n_i}\right)$$

$$\emptyset_{Fp} = \frac{kT}{e}\ln\left(\frac{N_a}{n_i}\right)$$

$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right)$$

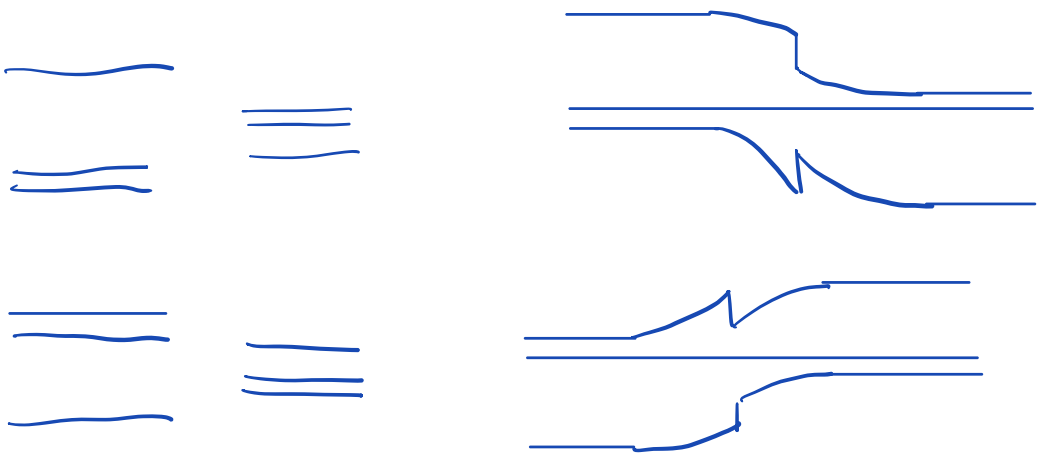


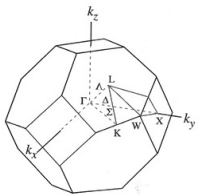
Example

Objective: Calculate the built-in potential barrier in a pn junction.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 2 \times 10^{17}\ \text{cm}^{-3}$ and $N_d = 10^{15}\ \text{cm}^{-3}$.

Assume: $n_i = 1.5 \times 10^{10}\text{cm}^{-3}$.



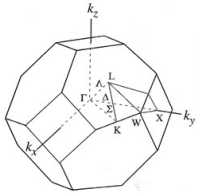


Drawing Band-Diagrams

- Homojunction

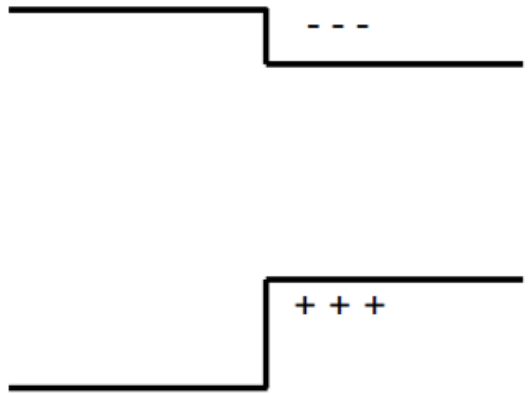


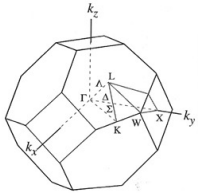




Drawing Band-Diagrams

- Heterojunction
  - AlGaAs/GaAs
  - InP/InGaAs





Drawing Band-Diagrams

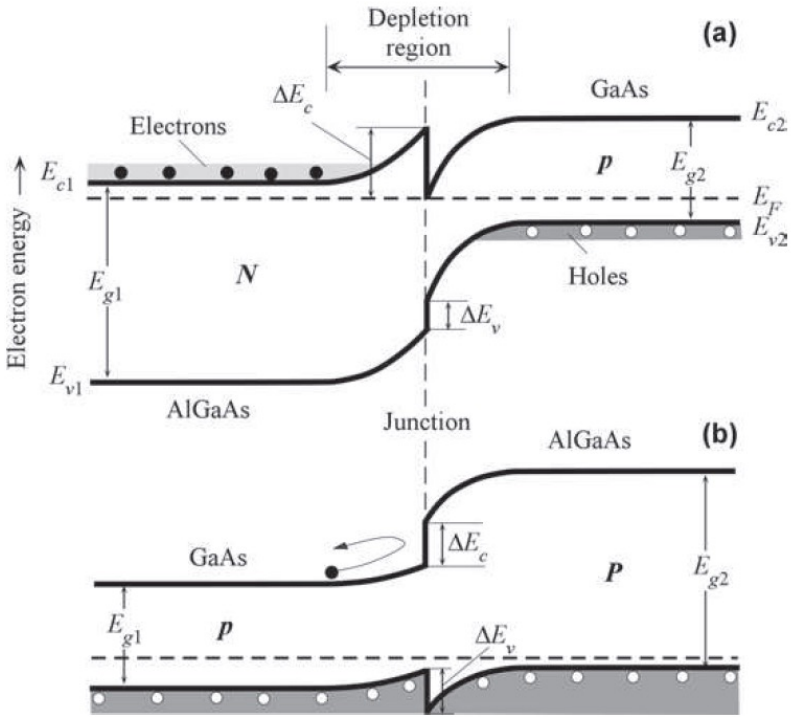
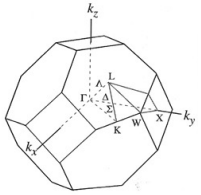


FIGURE 3.28 (a) *nP* and (b) *pP* heterojunctions and their energy band diagrams (schematic only to illustrate general features). Under open circuit and equilibrium conditions, the Fermi level E_F must be uniform, that is, continuous throughout the device. If E_F is close to the conduction band edge, E_c, it results in an *n*-type, and if it is close to the valence band edge, E_v, it results in a *p*-type semiconductor. There is a discontinuity ΔE_c as in E_c, and ΔE_v in E_v, right at the junction.



How to Calculate the Potential vs. Distance?

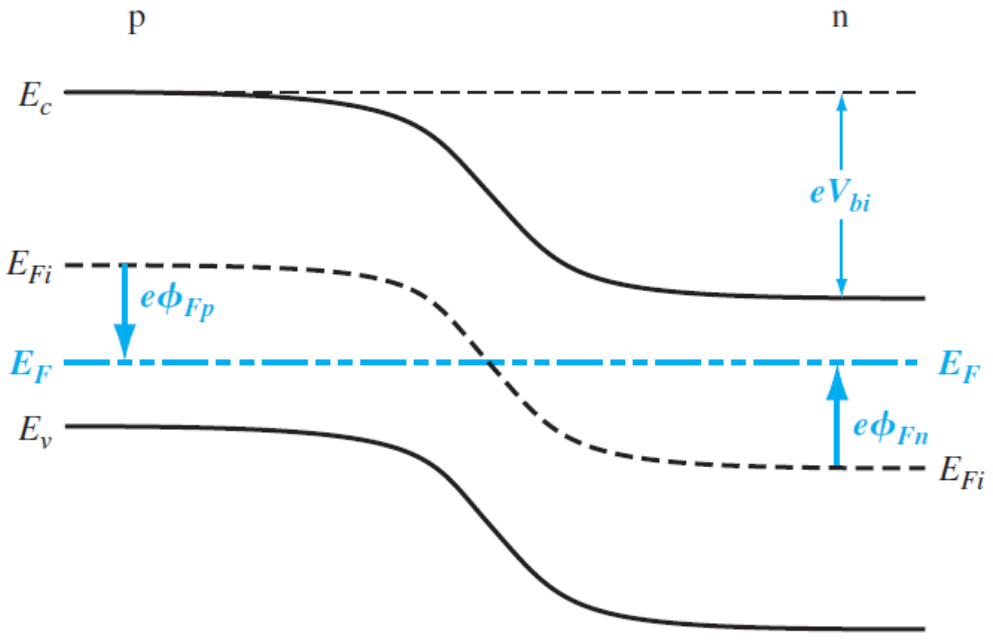
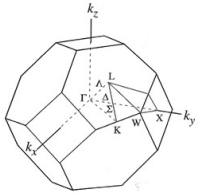


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.



Space Charge Region

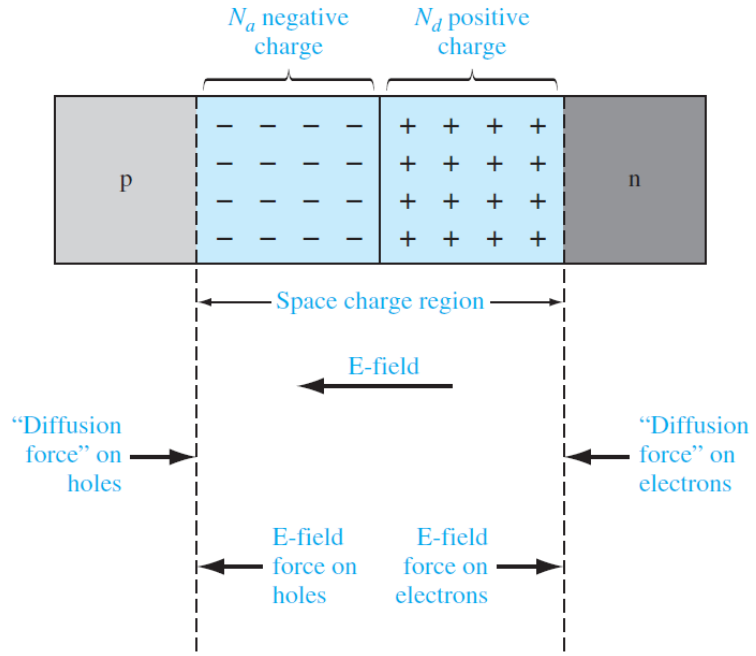
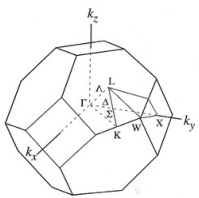
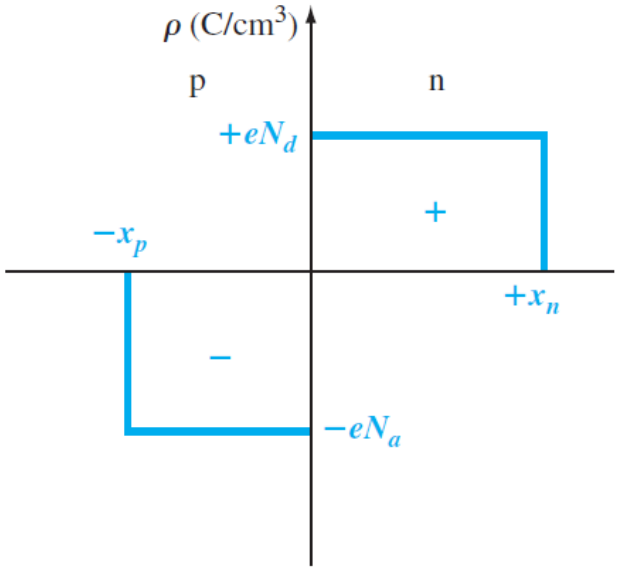


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

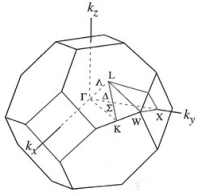


Space Charge Density



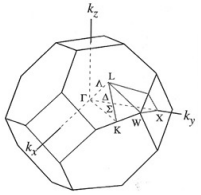
$$N_d x_n = N_a x_p$$

Figure 7.4 | The space charge density in a uniformly doped pn junction assuming the abrupt junction approximation.



Electric Field

- Poisson's Equation
  - $\frac{d^2\emptyset(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{dE(x)}{dx}$
  - $\rho(x) = \begin{cases} -eN_a & -x_p < x < 0 \\ eN_d & 0 < x < x_n \end{cases}$
  - $E(x) = \begin{cases} \frac{-eN_a}{\epsilon_s}(x + x_p) & -x_p < x < 0 \\ \frac{-eN_d}{\epsilon_s}(-x + x_n) & 0 < x < x_n \end{cases}$



Electric Field

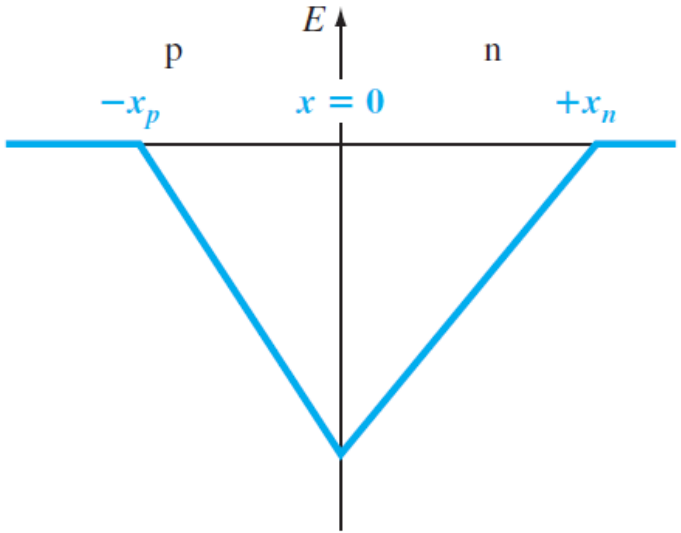
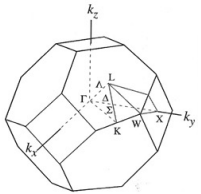


Figure 7.5 | Electric field in the space charge region of a uniformly doped pn junction.



Electric Potential

- $\emptyset(x) = -\int E(x)dx$

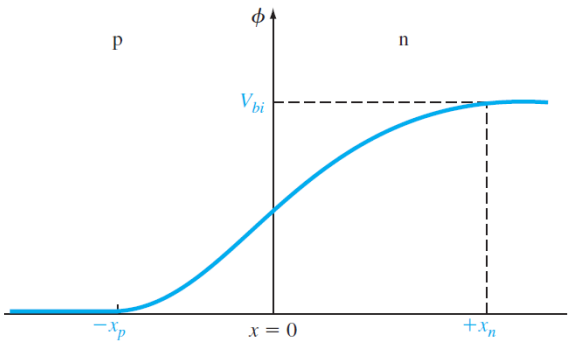
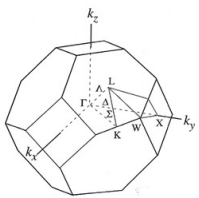


Figure 7.6 | Electric potential through the space charge region of a uniformly doped pn junction.



Electrical Potential

- The relation between electrical potential and electron potential energy:
  - $energy = -e\emptyset$

$$V_{bi} = |\emptyset(x = x_n)|$$

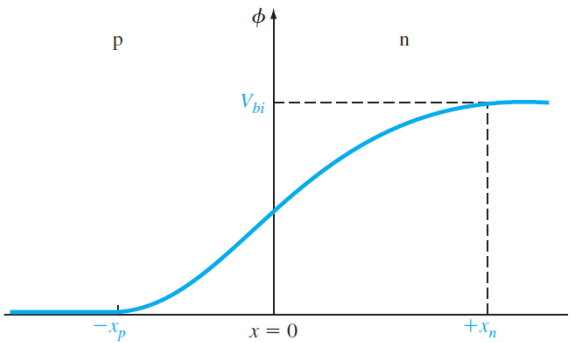


Figure 7.6 | Electric potential through the space charge region of a uniformly doped pn junction.

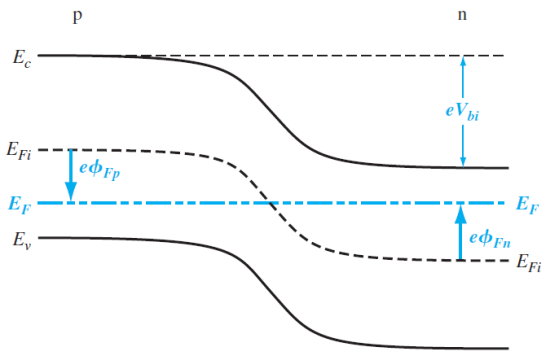
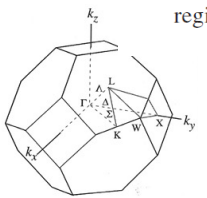
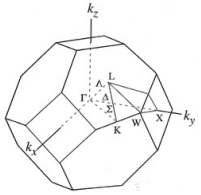


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.



Space Charge Width

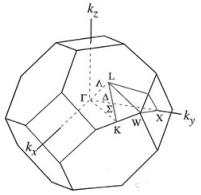
- $V_{bi} = \frac{kT}{e}\ln\left(\frac{N_a N_d}{n_i^2}\right)$
- $x_n = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_d}{N_a}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $W = x_p + x_n = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a+N_d}{N_a N_d}\right]\right\}^{1/2}$



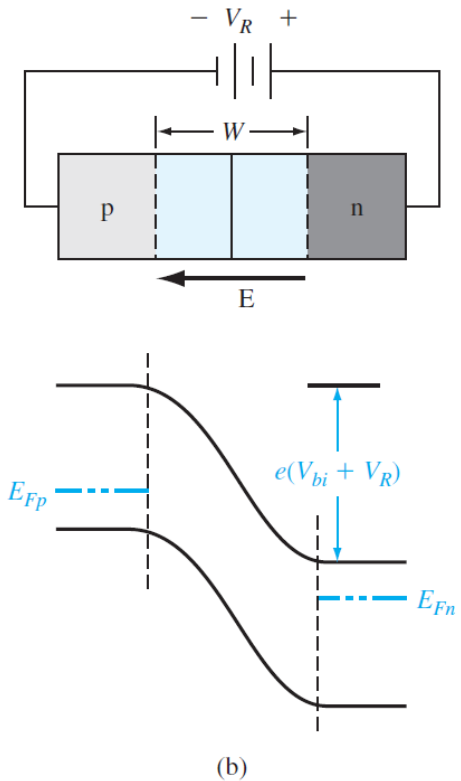
Example

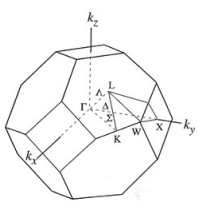
Objective: Calculate the space charge width and electric field in a pn junction for zero bias.

Consider a silicon pn junction at $T = 300$ K with doping concentrations of $N_a = 10^{16}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3}.



Reverse Applied Bias





Reverse Applied Bias

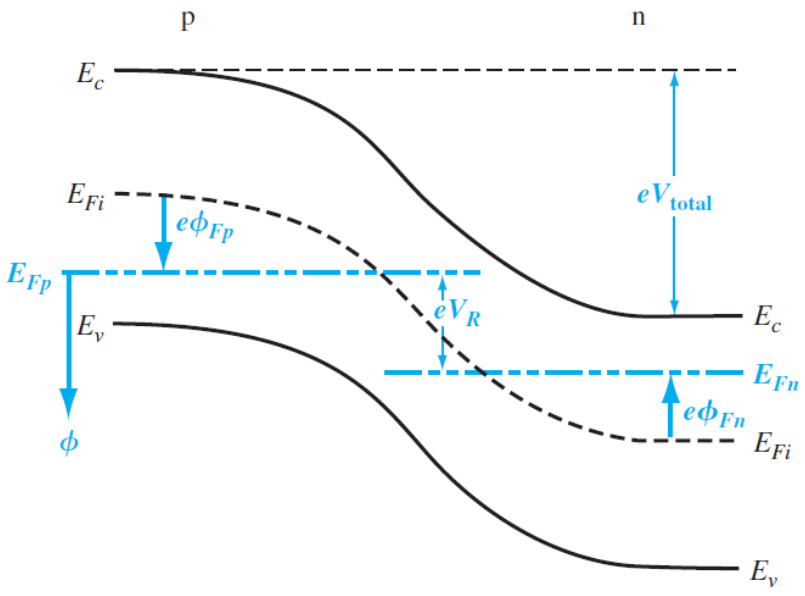
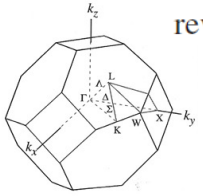


Figure 7.7 | Energy-band diagram of a pn junction under reverse bias.

- $V_{total} = |\emptyset_{Fn}| + |\emptyset_{Fp}| + |V_R|$
- $V_{total} = V_{bi} + |V_R|$
- $V_R = -V$

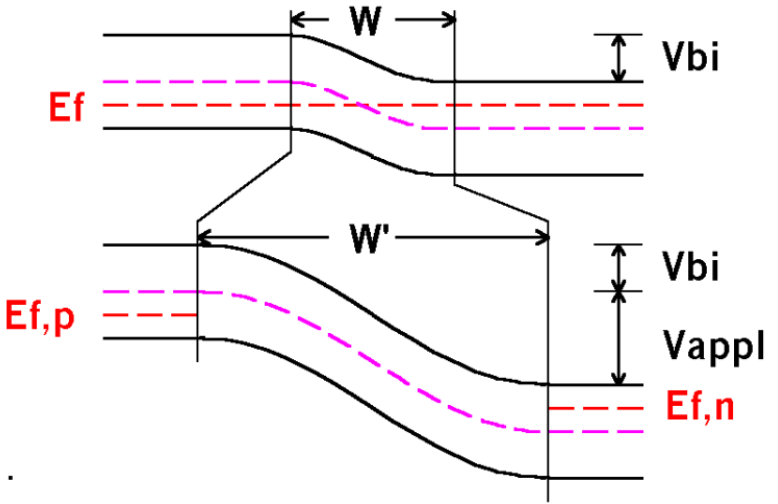
P区 负偏压

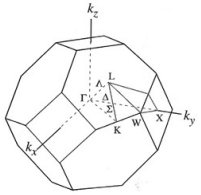
P区: $\varphi \downarrow$ $E = (-e)\varphi \uparrow$



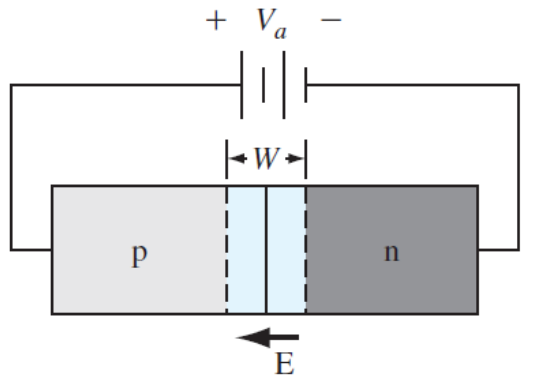
Total Space Charge Width with Reverse Bias

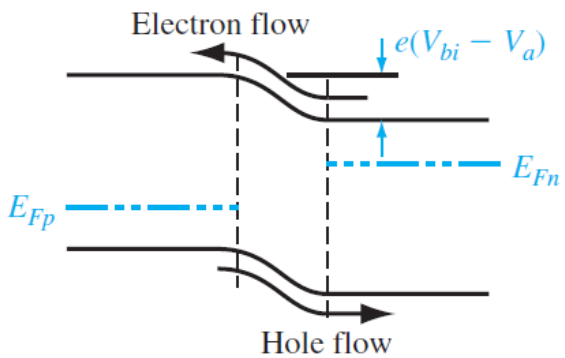
- $x_n = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $x_p = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_d}{N_a}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $W = x_p + x_n = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2}$
- $E_{max} = \frac{-2(V_{bi}-V)}{W}$

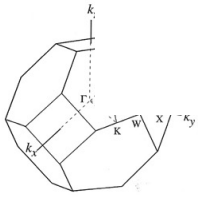




Forward Bias







- $x_n = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $x_p = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_d}{N_a}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$
- $W = x_p + x_n = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2}$
- $E_{max} = \frac{-2(V_{bi}-V)}{W}$

Junction Capacitance

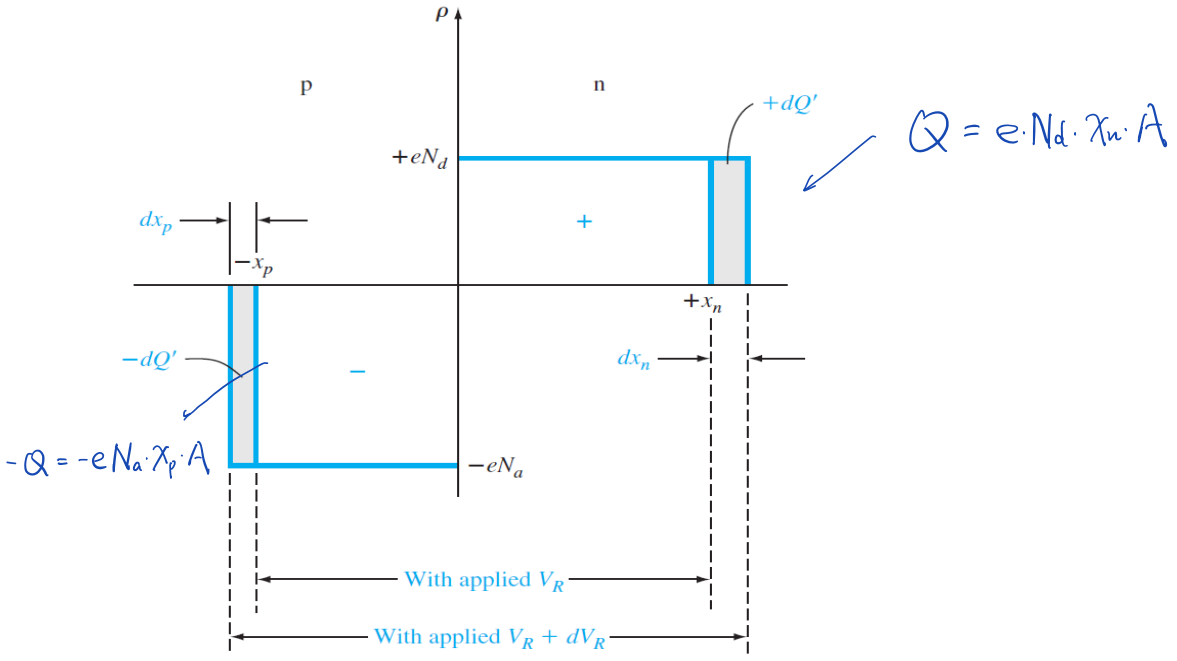
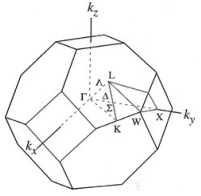


Figure 7.9 | Differential change in the space charge width with a differential change in reverse-biased voltage for a uniformly doped pn junction.



Junction Capacitance

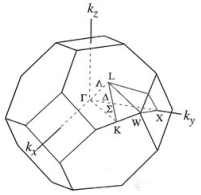
- $C' = \dfrac{dQ'}{dV_R}$
- $dQ' = eN_d dx_n = eN_a dx_p$
- $C' = \dfrac{dQ'}{dV_R} = eN_d \dfrac{dx_n}{dV_R}$
- $C' = \left\{ \dfrac{e\epsilon_s N_a N_d}{2(V_{bi}-V)(N_a+N_d)} \right\}^{1/2}$
- $C' = \dfrac{\epsilon_s}{W}$

$$\frac{dx_n}{dV} = \left(\frac{2\varepsilon_s}{e}\cdot\left(\frac{N_A}{N_D}\right)\cdot\left(\frac{1}{N_A+N_D}\right)\right)^{\frac{1}{2}}\cdot\frac{1}{2}\cdot\frac{1}{\sqrt{V_{bi}-V}}$$

$$C' = eN_d\cdot\frac{dx_n}{dV_R} = \sqrt{\frac{e\varepsilon_s\cdot N_a\cdot N_d}{2(V_{bi}-V)(N_a+N_d)}} = \frac{\varepsilon_s}{W}$$

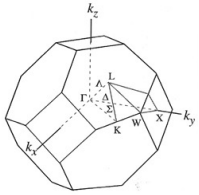
$$V_{bi}-V = \frac{1}{2}\left(\frac{eN_D}{\varepsilon}\cdot x_n^2 + \frac{eN_A}{\varepsilon}\cdot x_p^2\right)$$

$$-dV = \frac{eN_D}{\varepsilon}\cdot x_n\cdot dx_n + \frac{eN_A}{\varepsilon}\cdot x_p\, dx_p$$



One-Sides Junctions

- If $N_a \gg N_d$
- $W = x_p + x_n = \left\{\frac{2\epsilon_s(V_{bi}-V)}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2} \approx \left\{\frac{2\epsilon_s(V_{bi}-V)}{eN_d}\right\}^{1/2}$
- $C' = \left\{\frac{e\epsilon_s N_d}{2(V_{bi}-V)}\right\}^{1/2}$
- $\left(\frac{1}{C'}\right)^2 = \frac{2(V_{bi}-V)}{e\epsilon_s N_d}$



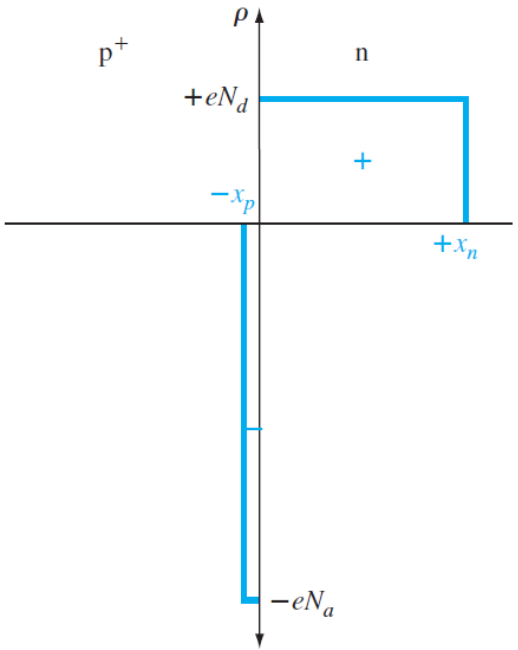


Figure 7.10 | Space charge density of a one-sided p⁻n junction.

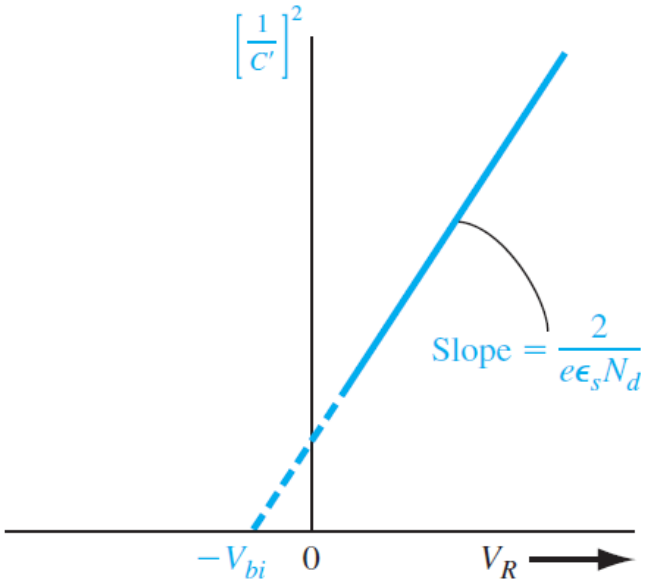


Figure 7.11 | $(1/C')^2$ versus V_R of a uniformly doped pn junction.

